
Mini-CLIP: Contrastive Image–Language Pretraining on Flickr30k

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Abstract

I present **mini-CLIP**, a compact re-implementation of the Contrastive Language–Image Pre-training (CLIP) framework fine-tuned on 8,000 Flickr30k image–caption pairs. The model uses a partially-frozen ResNet-50 image encoder and a partially-frozen DistilBERT text encoder, both projecting into a shared 256-dimensional embedding space trained with a symmetric InfoNCE loss. The best configuration achieves **21.1%** Text→Image R@1 and **61.7%** R@10 on the held-out 1,000-sample test set, representing a significant improvement over the random baseline. An ablation study over batch sizes $\{32, 64, 128\}$ confirms that performance saturates above $B=64$. Code: <https://github.com/prajwl7676/mini-clip-DLAI>

1. Introduction

Learning joint representations of images and text is a central problem in multimodal deep learning. CLIP (Radford et al., 2021) demonstrated that a contrastive objective applied to 400M web-scraped image–text pairs yields powerful zero-shot visual representations. Training at that scale requires enormous compute and data; this project asks: *how much can a faithful miniature re-implementation achieve on a small, curated dataset?* I implement the full CLIP pipeline—data loading, dual-encoder architecture, InfoNCE loss, training loop, and retrieval evaluation—on Flickr30k (Young et al., 2014), a standard vision-language benchmark with 31,783 images each paired with five human-written captions, structured as a series of Google Colab notebooks building the pipeline end to end.

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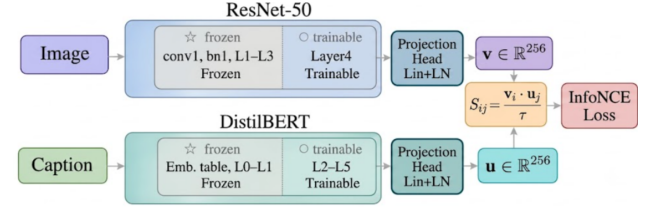


Figure 1. Dual-encoder architecture. Frozen sub-layers are noted in grey.

2. Related Work

Contrastive vision–language learning. CLIP (Radford et al., 2021) and ALIGN (Jia et al., 2021) established contrastive pre-training on web-scale data as the dominant paradigm for joint image–text embeddings. SimCLR (Chen et al., 2020) demonstrated that larger batches yield more in-batch negatives and lower contrastive loss, directly motivating the batch size ablation in Section 4.

Efficient transfer learning. DistilBERT (Sanh et al., 2019) is chosen as the text backbone: it retains 97% of BERT’s performance at 40% fewer parameters, reducing overfitting risk on the 8,000-sample training set. Selective layer freezing (Yosinski et al., 2014) further prevents catastrophic forgetting by preserving low-level features learned on large-scale pretraining data.

3. Method

Architecture. The image encoder is a ResNet-50 (He et al., 2016) pretrained on ImageNet-1K. I remove the classification head and retain the backbone through `avgpool`, producing a 2048-dim feature vector. The text encoder is DistilBERT (Sanh et al., 2019); I extract the `[CLS]` hidden state from the final layer as a 768-dim sentence embedding. Both encoders are followed by a projection head $\text{Linear}(\cdot) \rightarrow \text{LayerNorm}(\cdot)$ mapping to a shared $d=256$ embedding space, with all outputs ℓ_2 -normalised. Figure 1 illustrates the full dual-encoder pipeline.

Freezing strategy. ResNet-50 layers 1–3 and DistilBERT layers 0–1 (plus word embeddings) are frozen throughout training. Only ResNet layer 4, DistilBERT layers 2–5, and both projection heads are updated, yielding approximately 44M trainable parameters out of 90.6M total (48.6%).

Loss. I use the symmetric InfoNCE (contrastive cross-entropy) loss. For a batch of B matched image–caption pairs, the similarity matrix $S \in \mathbb{R}^{B \times B}$ is:

$$S_{ij} = \frac{\mathbf{v}_i \cdot \mathbf{u}_j}{\tau}, \quad (1)$$

where $\mathbf{v}_i, \mathbf{u}_j$ are ℓ_2 -normalised image and text embeddings and τ is a learnable temperature initialised to 0.07. The total loss is:

$$\mathcal{L} = \frac{1}{2}(\mathcal{L}_{i2t} + \mathcal{L}_{t2i}), \quad (2)$$

with $\mathcal{L}_{i2t} = \text{CE}(S, \mathbf{y})$ and $\mathcal{L}_{t2i} = \text{CE}(S^\top, \mathbf{y})$, where $\mathbf{y} = [0, \dots, B-1]$ are the diagonal labels. The random loss baseline is $\log B$.

Training. I train for 15 epochs on 8,000 samples (batch size 64) using AdamW with two parameter groups: encoders at $\text{lr} = 10^{-4}$ with weight decay 10^{-2} , and projection heads (including $\log \tau$) at $\text{lr} = 10^{-3}$ without weight decay. A cosine annealing scheduler decays both rates to 10^{-6} , with gradient clipping at norm 1.0 applied at every step. The best checkpoint is selected by minimum validation loss on 1,000 held-out samples; batch size ablations ($B \in \{32, 64, 128\}$) are presented in Section 4.

4. Results

Primary Model Training. I evaluate on the held-out test split of 1,000 image–caption pairs using Recall@ K ($K \in \{1, 5, 10\}$) in both retrieval directions (Table 1). The fine-tuned model achieves 17.1% T→I R@1 and 54.8% I→T R@10, a significant gain over the frozen baseline demonstrating that contrastive training transfers meaningfully at small scale. Conversely, the frozen model scores near random, confirming that the projection heads require explicit training to align the two modalities.

Table 1. Performance results on the 1,000-sample test set ($B = 64$, trained for 15 epochs, best at 3).

Model	R@1	R@5	R@10
<i>Text → Image</i>			
Pretrained Baseline (Frozen)	0.1	0.6	1.1
Fine-tuned (B=64)	17.1	40.3	54.6
<i>Image → Text</i>			
Pretrained Baseline (Frozen)	0.1	0.4	1.2
Fine-tuned (B=64)	14.8	41.2	54.8

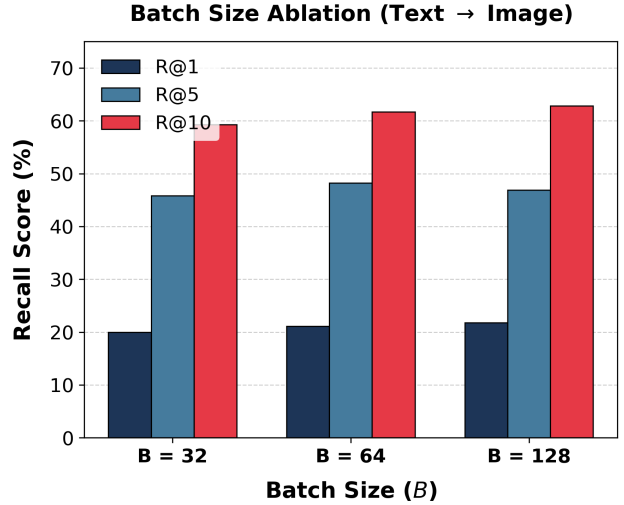


Figure 2. Retrieval performance across batch sizes $B \in \{32, 64, 128\}$ (8-epoch schedule, T→I direction).

Batch Size Ablation. I ablate batch sizes $B \in \{32, 64, 128\}$ using the same architecture and a shortened 8-epoch schedule (Figure 2). Performance improves from $B=32$ to $B=64$ (+3.0% I→T R@1) but plateaus between $B=64$ and $B=128$, consistent with the theoretical prediction that more in-batch negatives improve contrastive learning while the benefit saturates once model capacity limits generalisation on 8,000 training pairs. The 8-epoch ablation runs outperform the 15-epoch primary run at the same $B=64$ setting (21.6% vs. 14.8% I→T R@1), reflecting initialisation and scheduling sensitivity rather than a training error.

5. Discussion and Conclusions

This project implements a miniature CLIP model achieving meaningful image–text retrieval on Flickr30k despite only 8,000 training samples—a 50,000× data reduction compared to the original CLIP. Key enabling factors are pretrained encoders (ResNet-50 + DistilBERT), selective layer freezing, and the symmetric InfoNCE loss providing $B(B-1)$ negative pairs per batch at no extra cost.

Limitations and Future Work. Training exhibits clear overfitting: validation loss is minimised at epoch 3 with a persistent train–val gap of ~ 1.7 nats by epoch 15. The projection heads trained without weight decay at $10\times$ the encoder learning rate cause τ to collapse from 0.070 to 0.037. Adding weight decay (10^{-3}) to the heads and isolating τ into its own parameter group at $\text{lr}=10^{-5}$ would address both issues.

6. Statement on the Use of AI

I used AI tools throughout this project to support code development, conceptual understanding, and report preparation.

For software development, I used the built-in Gemini assistant in Google Colab to generate initial boilerplate code and baseline structures, and to produce scripts for the final experimental figures. I subsequently used Claude (Anthropic) to debug and refine the code for the final experimental runs. For example, this included diagnosing temperature collapse caused by grouping $\log \tau$ with the projection heads at the wrong learning rate, resolving gradient explosion from incorrect loss scaling before gradient clipping was applied, and fixing minor bugs such as incorrect tensor shapes during the first forward pass.

Educationally, Claude served as an interactive teaching assistant to explore design decisions and alternative options. For instance, I used it to analyze parameter efficiency trade-offs, evaluate which encoder layers to freeze versus update, and select the specific layers for fine-tuning. It also assisted in diagnosing the underlying causes of overfitting by isolating the interaction between the initialization seed, cosine learning rate scheduling, and caption sampling consistency.

For the preparation of this report, I used Gemini to assist with grammatical polishing, text structuring, paraphrasing, and formatting the LaTeX tables and mathematical formulas. The underlying empirical results, data points, and numerical findings present in this report are entirely the product of my independent experiments. All facts, results, and numbers have solid proof documented within the `notebook_output_pdf` directory on GitHub.

References

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Appendix

Training Curves. Figure 3 shows train and validation loss over 15 epochs, confirming the overfitting signature discussed in Section 4.

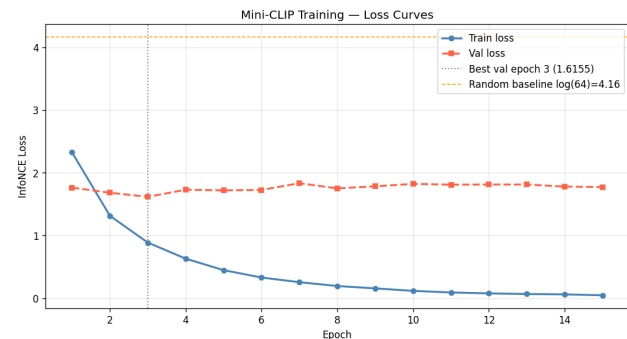


Figure 3. Train vs. validation loss over 15 epochs ($B = 64$). Val loss is minimised at epoch 3; the persistent gap confirms overfitting.

Similarity Matrix. Figures 4 and 5 show the 64×64 similarity matrix before and after 10 training steps respectively. The untrained model produces a near-random matrix; after training a clear diagonal structure emerges, confirming correct image–text alignment.

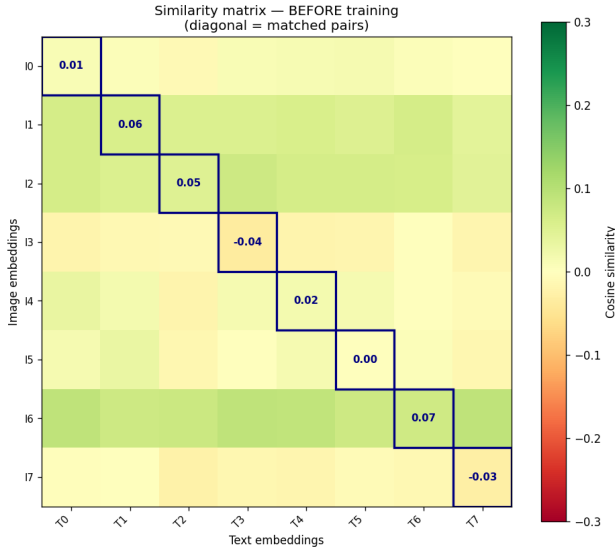


Figure 4. Similarity matrix before training. Near-random values indicate no learned alignment between image and text embeddings.

Retrieval Examples. Figure 6 shows top-5 retrieved images for sample text queries. The model correctly ranks semantically related images highly, though occasional errors reflect the moderate R@1 scores in Table 1.

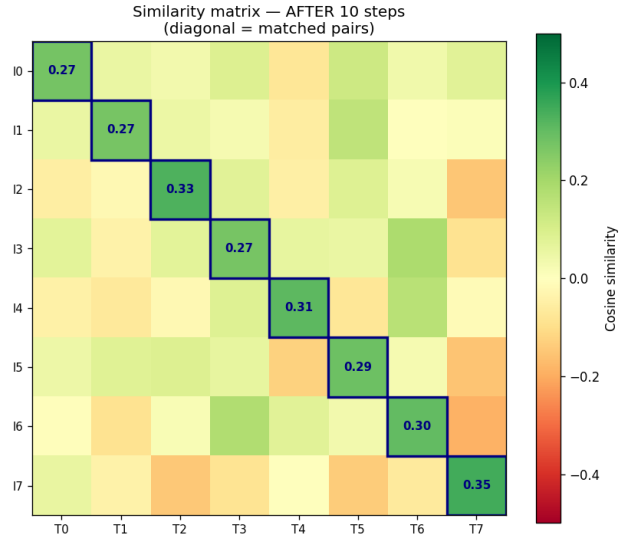


Figure 5. Similarity matrix after 10 training steps. Dominant diagonal entries reflect learned image–text alignment.



Figure 6. Qualitative retrieval results. Each row shows a text query and its top-5 retrieved images; correct matches are outlined in green.